

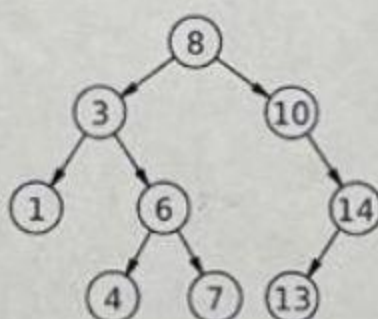
Exam, Data Structures

Solve as many of the following problems as you can. All questions are weighed equally. Be concise and to the point (irrelevant answers will **not** help towards partial credit). As a general rule, you should not spend (assuming you know the answer perfectly) more than five minutes on any single question (fifteen on the problems in the second part). Starting grade: **1 point**.

The time allotted for the exam is **one hour and 40 minutes**. The exam is **closed book**. Cheating attempts lead to failing the course.

1 For grades up to 6 (level one)

1. Write a function (or set of functions) to insert a node at the front of a *doubly linked list*. **Explain what you do.**
2. Consider a hashing scheme with open addressing that uses double hashing with a table size of 7, the first function $h_1(k) = k \pmod{7}$, and the second function $h_2(k) = 1 + k \pmod{5}$. We insert the elements 27, 43, 692, 72 successively into the table. Draw the hash table at the end of the process, explaining how insertion works.
3. Describe (giving a concrete example) how does delete by copy work.
4. We traverse the following binary tree.



Which sequences correspond to which traversals? Answers are 0.25 points each. You must get at least three of them correct, otherwise you get 0.

- a. 8,3,10,1,6,14,4,7,13.
- b. 1,3,4,6,7,8,10,13,14.
- c. 8,3,1,6,4,7,10,14,13.
- d. 1,4,7,6,3,13,14,10,8.

Answers:

- preorder: ...
- inorder: ...
- breadth-first search: ...
- postorder: ...

5. Draw the red-black trees that result after successively inserting the keys 42,39,32,13,20,9 into an initially empty red-black tree.

2 For higher grades (level two)

You must have a minimum of 5.5 on the first level for the second level to count.

6. Show how to determine whether a directed graph G contains a *universal sink*: a vertex with in-degree $n - 1$ and out-degree 0, in time $O(n)$, given an adjacency matrix for G .
7. Describe the idea of an algorithm that takes a max-heap (represented as a vector) and converts it to a min-heap. What is the complexity of the algorithm?
8. Give an algorithm with $O(n)$ time complexity and $O(n)$ space complexity that receives a binary tree and outputs the size of the largest subtree of T that is a binary search tree. **Explain why the algorithm has the desired complexities.**
9. An alternative method of performing an inorder tree walk of an n -node binary search tree finds the minimum element in the tree by calling TREE-MINIMUM and then making $n - 1$ calls to TREE-SUCCESSOR. Prove that this algorithm runs in $\Theta(n)$ time.